# Mark James Adams

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### Contributions to the generation of new ideas, tools, methodologies, or knowledge

- Conducted landmark ($N > 5$ million) genome-wide association study of depression, improving prediction accuracy and identifying mechanisms and drug targets.
- Develops computational pipelines for reproducible genomics and biomarker analyses.
- Combined psychometric and genetic modelling to understand depression symptoms and primate personality.
- Applied indirect genetic effects models to multilevel systems in behavioural ecology.

### The development of others and maintenance of effective working relationships

- Coordinates work among dozens of analysts in an international consortium.
- Co-leads special interest group in depression heterogeneity.
- Mentored research students in the creation of a peer support group for coding and data analysis.

### Contributions to the wider research and innovation community and to societal benefit

- Established and maintains open source repository for the Psychiatric Genomics Consortium.
- Engages regularly with lived-experience advisors.
- Facilites researcher access to international consortium datasets.

## Employment

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| 2024– | Senior Bioinformatician, Generation Scotland |
| 2021– | Senior Research Fellow, Centre for Clinical Brain Sciences, University of Edinburgh |
| 2017–2021 | Data Scientist, Division of Psychiatry, University of Edinburgh |
| 2015–2017 | Post-doctoral Research Fellow, Division of Psychiatry, University of Edinburgh |
| 2012–2015 | Post-doctoral Research Associate, Department of Animal and Plant Sciences, University of Sheffield |
| 2006–2007 | Laboratory Technician, Drosophila Genetic Resource Center, Kyoto Institute of Technology |

## Academic Service

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| 2023– | Academic Postgraduate Coordinator, Centre for Clinical Brain Sciences, University of Edinburgh |

## Education

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| 2008–2012 | PhD Psychology, University of Edinburgh |
| 2007–2008 | MSc Quantitative Genetics & Genome Analysis, University of Edinburgh |
| 1999–2003 | AB Theoretical Evolutionary Biology, Cornell University |

## External Roles

### Psychiatric Genomics Consortium

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| 2025– | Core Analytical Group Co-Director, Major Depressive Disorder Working Group |
| 2019– | Open Science Maintainer |
| 2019–2025 | Stage 2 Analyst, Major Depressive Disorder Working Group |
| 2017–2019 | Data Access Committee Representative, Major Depressive Disorder Working Group |

## Grants

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| 2025–2029 | Wellcome Trust. *What is the molecular genetic basis of anxiety in a very-low-income African setting? The Genetics of Anxiety In Malawi (G-AIM) Study.* (309744/Z/24/Z) £3,976,039. (Co-I) |
| 2025–2026 | BHF REA4 Pump Priming Award. *Multi-ancestry finemapping of gene clusters implicated in cardiovascular disease and mental health.* (RE/24/130012). £3500 |
| 2021–2024 | Medical Research Council. *DATAMIND: Data Hub for Mental health INformatics research Development.* £225,575. (Co-I) |
| 2014 | Wellcome Trust. *Inferring genetic architecture from high-density genomic data.* £2000 (Co-I) |
| 2010 | Moray Endowment Fund Award. *Selection on human personality.* £2280. |
| 2009 | Sasakawa Foundation. *Personality in Japanese macaques.* £1000. |

## Awards

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| 2025 | Gershon Paper of the Year Award, International Society of Psychiatric Genetics. For: Adams, Streit, et al., 2025. *(Given to a paper which has been published and reported on an outstanding achievement in the field of psychiatric genetics.)* |
| 2012 | Fulker Award, Behavior Genetics Association. For: Adams et al., 2012. *(Given to a paper of unusual merit that appeared in Behavior Genetics.)* |
| 2008 | Sir Kenneth Mather Memorial Prize, Genetics Society. For: MSc thesis. *(Outstanding performance in the area of quantitative genetics.)* |

## Team Recognition

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| 2025 | Team Excellence Award, College of Medicine and Veterinary Medicine, University of Edinburgh. For: Generation Scotland. |
| 2022 | Good Research Practice Award, University of Edinburgh. For: Depression Detectives *(2nd Prize in 'Responsible Research' category.)* |

## Publications

### As Lead Author

*Studies designed and lead.*

Chuong, M., Kwong, A. S. F., Amador, C., Haley, C. S., McIntosh, A. M., & Adams, M. J. (2026). Methodological considerations when using polygenic scores to explore parent-offspring genetic nurturing effects. *Wellcome Open Research*, *11*(19). https://doi.org/10.12688/wellcomeopenres.24955.1

Adams, M. J., Streit, F., Meng, X., Awasthi, S., Adey, B. N., Choi, K. W., Chundru, V. K., Coleman, J. R. I., Ferwerda, B., Foo, J. C., Gerring, Z. F., Giannakopoulou, O., Gupta, P., Hall, A. S. M., Harder, A., Howard, D. M., Hübel, C., Kwong, A. S. F., Levey, D. F., … McIntosh, A. M. (2025). Trans-ancestry genome-wide study of depression identifies 697 associations implicating cell types and pharmacotherapies. *Cell*, *188*(3), 640–652. https://doi.org/10.1016/j.cell.2024.12.002

Adams, M. J., Thorp, J. G., Jermy, B. S., Kwong, A. S. F., Kõiv, K., Grotzinger, A. D., Nivard, M. G., Marshall, S., Milaneschi, Y., Baune, B. T., & et al. (2024). Genome-wide meta-analysis of ascertainment and symptom structures of major depression in case-enriched and community cohorts. *Psychological Medicine*, *54*(12), 3459–3468. https://doi.org/10.1017/S0033291724001880

Adams, M., Hill, W., Howard, D., Dashti, H., Davis, K., Campbell, A., Clarke, T.-K., Deary, I., Hayward, C., Porteous, D., Hotopf, M., & McIntosh, A. (2020). Factors associated with sharing e-mail information and mental health survey participation in large population cohorts. *International Journal of Epidemiology*, *49*(2), 410–421. https://doi.org/10.1093/ije/dyz134

Adams, M., Howard, D., Luciano, M., Clarke, T.-K., Davies, G., Hill, W., Smith, D., Deary, I., Porteous, D., McIntosh, A., 23andMe Research Team, & Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2020). Genetic stratification of depression by neuroticism: Revisiting a diagnostic tradition. *Psychological Medicine*, *50*(15), 2526–2535. https://doi.org/10.1017/S0033291719002629

Arden, R., & Adams, M. (2016). A general intelligence factor in dogs. *Intelligence*, *55*, 79–85. https://doi.org/10.1016/j.intell.2016.01.008

Adams, M., Bonaventura, M., Ostner, J., Schulke, O., De Marco, A., Thierry, B., Engelhardt, A., Widdig, A., Gerald, M., & Weiss, A. (2015). Personality structure and social style in macaques. *Journal of Personality and Social Psychology*, *109*(2), 338–353. https://doi.org/10.1037/pspp0000041

Adams, M., Robinson, M., Mannarelli, M., & Hatchwell, B. (2015a). Social genetic and social environment effects on parental and helper care in a cooperatively breeding bird. *Proceedings of the Royal Society B-Biological Sciences*, *282*(1810). https://doi.org/10.1098/rspb.2015.0689

Hatchwell, B., Gullett, P., & Adams, M. (2014). Helping in cooperatively breeding long-tailed tits: A test of Hamilton's rule. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *369*(1642). https://doi.org/10.1098/rstb.2013.0565

Adams, M., King, J., & Weiss, A. (2012). The majority of genetic variation in orangutan personality and subjective well-being is nonadditive. *Behavior Genetics*, *42*, 675–686. https://doi.org/10.1007/s10519-012-9537-y

### Reviews & Commentary

Adams, M. J. (2023). Leveraging family relatedness to detect participation bias in genetic studies. *Nature Genetics*. https://doi.org/10.1038/s41588-023-01440-9

Arden, R., Bensky, M., & Adams, M. (2016). A review of cognitive abilities in dogs, 1911 through 2016: More individual differences, please! *Current Directions in Psychological Science*, *25*(5), 307–312. https://doi.org/10.1177/0963721416667718

Adams, M. (2014). Feasibility and uncertainty in behavior genetics for the nonhuman primate. *International Journal of Primatology*, *35*(1), 156–168. https://doi.org/10.1007/s10764-013-9722-8

### As Main Author

*Provided primary data, analysis, and design.*

Casey, H., Wright, H., King, J., Adams, M., McIntosh, A., Strawbridge, R., Fallon, M., Smith, D., Murray, A., & Whalley, H. (2025). Impact of lifestyle in chronic pain and depression: A propensity score analysis in UK Biobank. *Wellcome Open Research*, *10*(318). https://doi.org/10.12688/wellcomeopenres.23982.1

Davis, K. A., Coleman, J. R., Adams, M., Breen, G., Cai, N., Davies, H. L., Davies, K., Dregan, A., Eley, T. C., Fox, E., Holliday, J., Hübel, C., John, A., Kassam, A. S., Kempton, M. J., Lee, W., Li, D., Maina, J., McCabe, R., . . . Hotopf, M. (2025). The UK Biobank mental health enhancement 2022: Methods and results. *PLOS ONE*, *20*(5), 1–20. https://doi.org/10.1371/journal.pone.0324189

Davyson, E., Shen, X., Huider, F., Adams, M. J., Borges, K., McCartney, D. L., Barker, L. F., van Dongen, J., Boomsma, D. I., Weihs, A., Grabe, H. J., Kühn, L., Teumer, A., Völzke, H., Zhu, T., Kaprio, J., Ollikainen, M., David, F. S., Meinert, S., . . . McIntosh, A. M. (2025). Insights from a methylome-wide association study of antidepressant exposure. *Nature Communications*, *16*(1), 1908. https://doi.org/10.1038/s41467-024-55356-x

Edmondson-Stait, A. J., Davyson, E., Shen, X., Adams, M. J., Khandaker, G. M., Miron, V. E., McIntosh, A. M., Lawrie, S. M., Kwong, A. S., & Whalley, H. C. (2025). Associations between IL-6 and trajectories of depressive symptoms across the life course: Evidence from ALSPAC and UK Biobank cohorts. *European Psychiatry*, *68*(1), e27. https://doi.org/10.1192/j.eurpsy.2025.7

Kanjira, S. C., Adams, M. J., Jiang, Y., Tian, C., 23andMe Research Team, Lewis, C. M., Kuchenbaecker, K., & McIntosh, A. M. (2025). Polygenic prediction of major depressive disorder and related traits in African ancestries UK Biobank participants. *Molecular Psychiatry*, *30*(1), 151–157. https://doi.org/10.1038/s41380-024-02662-x

Shen, X., Barbu, M., Caramaschi, D., Arathimos, R., Czamara, D., David, F. S., Dearman, A., Dilkes, E., Herrera-Rivero, M., Huider, F., Kühn, L., Lu, K.-C., Palviainen, T., Schowe, A. M., Shireby, G., Weihs, A., Wong, C. C. Y., Davyson, E., Casey, H., . . . McIntosh, A. M. (2025). A methylome-wide association study of major depression with out-of-sample case–control classification and trans-ancestry comparison. *Nature Mental Health*, *3*(10), 1152–1167. https://doi.org/10.1038/s44220-025-00486-4

Shen, X., Toenders, Y. J., Han, L. K. M., Weihs, A., Alexander, N., Andlauer, T. F. M., Brosch, K., Forstner, A. J., Grotegerd, D., Hahn, T., Hermesdorf, M., Hosten, N., Jamalabadi, H., Meinert, S., Milaneschi, Y., Sämann, P. G., Stein, F., Stolicyn, A., Teutenberg, L., … Whalley, H. C. (2025). Association between polygenic risk for major depression and brain structure in a mega-analysis of 50,975 participants across 11 studies. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-025-03136-4

Thomas, J. T., Thorp, J. G., Huider, F., Grimes, P. Z., Wang, R., Youssef, P., Coleman, J. R. I., Byrne, E. M., Adams, M., BIONIC consortium, The GLAD Study, Medland, S. E., Hickie, I. B., Olsen, C. M., Whiteman, D. C., Whalley, H. C., Penninx, B. W. J. H., van Loo, H. M., Derks, E. M., … Mitchell, B. L. (2025). Sex-stratified genome-wide association meta-analysis of major depressive disorder. *Nature Communications*, *16*(1), 7960. https://doi.org/10.1038/s41467-025-63236-1

Grimes, P. Z., Adams, M. J., Thng, G., Edmonson-Stait, A. J., Lu, Y., McIntosh, A., Cullen, B., Larsson, H., Whalley, H. C., & Kwong, A. S. F. (2024). Genetic architectures of adolescent depression trajectories in 2 longitudinal population cohorts. *JAMA Psychiatry*. https://doi.org/10.1001/jamapsychiatry.2024.0983

Thng, G., Shen, X., Stolicyn, A., Adams, M. J., Yeung, H. W., Batziou, V., Conole, E. L. S., Buchanan, C. R., Lawrie, S. M., Bastin, M. E., & et al. (2024). A comprehensive hierarchical comparison of structural connectomes in major depressive disorder cases v. controls in two large population samples. *Psychological Medicine*, *54*(10), 2515–2526. https://doi.org/10.1017/S0033291724000643

Yeung, H. W., Stolicyn, A., Shen, X., Adams, M. J., Romaniuk, L., Thng, G., Buchanan, C. R., Tucker-Drob, E. M., Bastin, M. E., McIntosh, A. M., Cox, S. R., Smith, K. M., & Whalley, H. C. (2024). Classification accuracy of structural and functional connectomes across different depressive phenotypes. *Imaging Neuroscience*, *2*, 1–24. https://doi.org/10.1162/imag_a_00064

Davyson, E., Shen, X., Gadd, D. A., Bernabeu, E., Hillary, R. F., McCartney, D. L., Adams, M., Marioni, R., & McIntosh, A. M. (2023). Metabolomic investigation of major depressive disorder identifies a potentially causal association with polyunsaturated fatty acids. *Biological Psychiatry*. https://doi.org/10.1016/j.biopsych.2023.01.027

de Nooij, L., Adams, M. J., Hawkins, E. L., Romaniuk, L., Munafò, M. R., Penton-Voak, I. S., Elliott, R., Bland, A. R., Waiter, G. D., Sandu, A.-L., & et al. (2023). Associations of negative affective biases and depressive symptoms in a community-based sample. *Psychological Medicine*, *53*(12), 5518–5527. https://doi.org/10.1017/S0033291722002720

Gao, C., Amador, C., Walker, R. M., Campbell, A., Madden, R. A., Adams, M. J., Bai, X., Liu, Y., Li, M., Hayward, C., Porteous, D. J., Shen, X., Evans, K. L., Haley, C. S., McIntosh, A. M., Navarro, P., & Zeng, Y. (2023). Phenome-wide analyses identify an association between the parent-of-origin effects dependent methylome and the rate of aging in humans. *Genome Biology*, *24*(1), 117. https://doi.org/10.1186/s13059-023-02953-6

Stolicyn, A., Lyall, L. M., Lyall, D. M., Høier, N. K., Adams, M. J., Shen, X., Cole, J. H., McIntosh, A. M., Whalley, H. C., & Smith, D. J. (2023). Comprehensive assessment of sleep duration, insomnia, and brain structure within the UK Biobank cohort. *Sleep*, *47*(2), zsad274. https://doi.org/10.1093/sleep/zsad274

Wang, X., Walker, A., Revez, J. A., Ni, G., Adams, M. J., McIntosh, A. M., Wray, N. R., Ripke, S., Mattheisen, M., Trzaskowski, M., Byrne, E. M., Abdellaoui, A., Adams, M. J., Agerbo, E., Air, T. M., Andlauer, T. F. M., Bacanu, S.-A., Bækvad-Hansen, M., Beekman, A. T. F., … Wray, N. R. (2023). Polygenic risk prediction: Why and when out-of-sample prediction $R^2$ can exceed SNP-based heritability. *The American Journal of Human Genetics*, *110*(7), 1207–1215.

https://doi.org/10.1016/j.ajhg.2023.06.006
doi: 10.1016/j.ajhg.2023.06.006.

Barbu, M. C., Amador, C., Kwong, A. S., Shen, X., Adams, M. J., Howard, D. M., Walker, R. M., Morris, S. W., Min, J. L., Liu, C., van Dongen, J., Ghanbari, M., Relton, C., Porteous, D. J., Campbell, A., Evans, K. L., Whalley, H. C., & McIntosh, A. M. (2022). Complex trait methylation scores in the prediction of major depressive disorder. *eBioMedicine*, *79*, 104000. https://doi.org/10.1016/j.ebiom.2022.104000

Barbu, M. C., Harris, M., Shen, X., Aleks, S., Green, C., Amador, C., Walker, R., Morris, S., Adams, M., Sandu, A., McNeil, C., Waiter, G., Evans, K., Campbell, A., Wardlaw, J., Steele, D., Murray, A., Porteous, D., McIntosh, A., & Whalley, H. (2022). Epigenome-wide association study of global cortical volumes in Generation Scotland: Scottish Family Health Study. *Epigenetics*, *17*(10), 1143–1158. https://doi.org/10.1080/15592294.2021.1997404

Barbu, M., Huider, F., Campbell, A., Amador, C., Adams, M., Lynall, M.-E., Howard, D., Walker, R., Morris, S., van Dongen, J., Porteous, D., Evans, K., Bullmore, E., Willemsen, G., Boomsma, D., Whalley, H., & McIntosh, A. (2022). Methylome-wide association study of antidepressant use in Generation Scotland and the Netherlands Twin Register implicates the innate immune system. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-021-01412-7

Chuong, M., Adams, M. J., Kwong, A. S. F., Haley, C. S., Amador, C., & McIntosh, A. M. (2022). Genome-by-trauma exposure interactions in adults with depression in the UK Biobank. *JAMA Psychiatry*, *79*(11), 1110–1117. https://doi.org/10.1001/jamapsychiatry.2022.2983

Grotzinger, A. D., Mallard, T. T., Akingbuwa, W. A., Ip, H. F., Adams, M. J., Lewis, C. M., McIntosh, A. M., Grove, J., Dalsgaard, S., Lesch, K.-P., Strom, N., Meier, S. M., Mattheisen, M., Børglum, A. D., Mors, O., Breen, G., Meier, S. M., Lee, P. H., Kendler, K. S., … Schizophrenia Working Group of the Psychiatric Genetics Consortium. (2022). Genetic architecture of 11 major psychiatric disorders at biobehavioral, functional genomic and molecular genetic levels of analysis. *Nature Genetics*. https://doi.org/10.1038/s41588-022-01057-4

Shen, X., Caramaschi, D., Adams, M. J., Walker, R. M., Min, J. L., Kwong, A., Hemani, G., Barbu, M. C., Whalley, H. C., Harris, S. E., Deary, I. J., Morris, S. W., Cox, S. R., Relton, C. L., Marioni, R. E., Evans, K. L., McIntosh, A. M., & Genetics of DNA Methylation Consortium. (2022). DNA methylome-wide association study of genetic risk for depression implicates antigen processing and immune responses. *Genome Medicine*, *14*(1), 36. https://doi.org/10.1186/s13073-022-01039-5

Barbu, M., Harris, M., Shen, X., Stolicyn, A., Green, C., Amador, C., Walker, R., Morris, S., Adams, M., Sandu, A., McNeil, C., Waiter, G., Evans, K., Campbell, A., Wardlaw, J., Steele, D., Murray, A., Porteous, D., McIntosh, A., & Whalley, H. (2021). Epigenome-wide association study of global cortical volumes in Generation Scotland: Scottish Family Health Study. *Epigenetics*. https://doi.org/10.1080/15592294.2021.1997404

Howard, D., Pain, O., Arathimos, R., Barbu, M., Amador, C., Walker, R., Jermy, B., Adams, M., Deary, I., Porteous, D., Campbell, A., Sullivan, P., Evans, K., Arseneault, L., Wray, N., Meaney, M., McIntosh, A., & Lewis, C. (2021). Methylome-wide association study of early life stressors and adult mental health. *Human Molecular Genetics*. https://doi.org/10.1093/hmg/ddab274

Johnston, K., Ward, J., Ray, P., Adams, M., McIntosh, A., Smith, B., Strawbridge, R., Price, T., Smith, D., Nicholl, B., Bailey, M., & Epstein, M. (2021). Sex-stratified genome-wide association study of multisite chronic pain in UK Biobank. *PLoS Genetics*, *17*(4), e1009428. https://doi.org/10.1371/journal.pgen.1009428

Schoorl, J., Barbu, M., Shen, X., Harris, M., Adams, M., Whalley, H., & Lawrie, S. (2021). Grey and white matter associations of psychotic-like experiences in a general population sample UK Biobank. *Translational Psychiatry*, *11*(1). https://doi.org/10.1038/s41398-020-01131-7

Shen, X., MacSweeney, N., Chan, S., Barbu, M., Adams, M., Lawrie, S., Romaniuk, L., McIntosh, A., & Whalley, H. (2021). Brain structural associations with depression in a large early adolescent sample (the ABCD study®). *EClinicalMedicine*, *42*. https://doi.org/10.1016/j.eclinm.2021.101204

Van Dongen, J., Hagenbeek, F., Suderman, M., Roetman, P., Sugden, K., Chiocchetti, A., Ismail, K., Mulder, R., Hafferty, J., Adams, M., Walker, R., Morris, S., Lahti, J., Küpers, L., Escaramis, G., Alemany, S., Jan Bonder, M., Meijer, M., Ip, H., … Biobank-based Integrative Omics Study Consortium. (2021). DNA methylation signatures of aggression and closely related constructs: A meta-analysis of epigenome-wide studies across the lifespan. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-020-00987-x

Barbu, M., Spiliopoulou, A., Colombo, M., McKeigue, P., Clarke, T.-K., Howard, D., Adams, M., Shen, X., Lawrie, S., McIntosh, A., & Whalley, H. (2020). Expression quantitative trait loci-derived scores and white matter microstructure in UK Biobank: A novel approach to integrating genetics and neuroimaging. *Translational Psychiatry*, *10*. https://doi.org/10.1038/s41398-020-0724-y

Davis, K., Coleman, J., Adams, M., Allen, N., Breen, G., Cullen, B., Dickens, C., Fox, E., Graham, N., Holliday, J., Howard, L., John, A., Lee, W., Mccabe, R., McIntosh, A., Pearsall, R., Smith, D., Sudlow, C., Ward, J., … Hotopf, M. (2020). Mental health in UK Biobank – development, implementation and results from an online questionnaire completed by 157 366 participants: A reanalysis. *BJPsych Open*, *6*(2). https://doi.org/10.1192/bjo.2019.100

de Nooij, L., Harris, M., Adams, M., Clarke, T.-K., Shen, X., Cox, S., McIntosh, A., & Whalley, H. (2020). Cognitive functioning and lifetime major depressive disorder in UK Biobank. *European Psychiatry*, *63*(1). https://doi.org/10.1192/j.eurpsy.2020.24

Green, C., Shen, X., Stevenson, A., Conole, E., Harris, M., Barbu, M., Hawkins, E., Adams, M., Hillary, R., Lawrie, S., Evans, K., Walker, R., Morris, S., Porteous, D., Wardlaw, J., Douglas Steele, J., Waiter, G., Sandu, A.-L., Campbell, A., … Whalley, H. (2020). Structural brain correlates of serum and epigenetic markers of inflammation in major depressive disorder. *Brain, Behavior, and Immunity*. https://doi.org/10.1016/j.bbi.2020.11.024

Howard, D., Folkersen, L., Coleman, J., Adams, M., Glanville, K., Werge, T., Hagenaars, S., Han, B., Porteous, D., Campbell, A., Clarke, T.-K., Breen, G., Sullivan, P., Wray, N., Lewis, C., & McIntosh, A. (2020). Genetic stratification of depression in UK Biobank. *Translational Psychiatry*, *10*(1). https://doi.org/10.1038/s41398-020-0848-0

Kwong, A., Pearson, R., Adams, M., Northstone, K., Tilling, K., Smith, D., Fawns-ritchie, C., Bould, H., Warne, N., Zammit, S., Gunnell, D., Moran, P., Micali, N., Reichenberg, A., Hickman, M., Rai, D., Haworth, S., Campbell, A., Altschul, D., … Timpson, N. (2020). Mental health before and during COVID-19 in two longitudinal UK population cohorts. *The British Journal of Psychiatry*, 1–27. https://doi.org/10.1192/bjp.2020.242

Meng, W., Adams, M., Reel, P., Rajendrakumar, A., Huang, Y., Deary, I., Palmer, C., McIntosh, A., & Smith, B. (2020). Genetic correlations between pain phenotypes and depression and neuroticism. *European Journal of Human Genetics*, *28*, 358–366. https://doi.org/10.1038/s41431-019-0530-2

Meng, W., Chan, B., Harris, C., Freidin, M., Hebert, H., Adams, M., Campbell, A., Hayward, C., Zheng, H., Zhang, X., Colvin, L., Hales, T., Palmer, C., Williams, F., McIntosh, A., & Smith, B.

(2020). A genome-wide association study finds genetic variants associated with neck or shoulder pain in UK Biobank. *Human Molecular Genetics*. https://doi.org/10.1093/hmg/ddaa058

Shen, X., Howard, D., Adams, M., Hill, W., Clarke, T.-K., Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Deary, I., Whalley, H., & McIntosh, A. (2020). A phenome-wide association and Mendelian Randomisation study of polygenic risk for depression in UK Biobank. *Nature Communications*, *11*. https://doi.org/10.1038/s41467-020-16022-0

Stolicyn, A., Harris, M., Shen, X., Barbu, M., Adams, M., Hawkins, E., Nooij, L., Yeung, H., Murray, A., Lawrie, S., Steele, J., McIntosh, A., & Whalley, H. (2020). Automated classification of depression from structural brain measures across two independent community-based cohorts. *Human Brain Mapping*. https://doi.org/10.1002/hbm.25095

Arnau Soler, A., Adams, M., Clarke, T.-K., MacIntyre, D., Milburn, K., Navrady, L., Generation Scotland, Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Hayward, C., McIntosh, A., & Thomson, P. (2019). A validation of the diathesis-stress model for depression in Generation Scotland. *Translational Psychiatry*. https://doi.org/10.1038/s41398-018-0356-7

Arnau Soler, A., MacDonald-Dunlop, E., Adams, M., Clarke, T.-K., MacIntyre, D., Milburn, K., Navrady, L., Generation Scotland, Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Hayward, C., McIntosh, A., & Thomson, P. (2019). Genome-wide by environment interaction studies of depressive symptoms and psychosocial stress in UK Biobank and Generation Scotland. *Translational Psychiatry*, *9*(1). https://doi.org/10.1038/s41398-018-0360-y

Barbu, M., Zeng, Y., Shen, X., Cox, S., Clarke, T.-K., Gibson, J., Adams, M., Johnstone, M., Haley, C., Lawrie, S., Deary, I., McIntosh, A., & Whalley, H. (2019). Association of whole-genome and NETRIN1 signaling pathway-derived polygenic risk scores for major depressive disorder and white matter microstructure in UK Biobank. *Biological psychiatry. Cognitive neuroscience and neuroimaging*, *4*(1), 91–100. https://doi.org/10.1016/j.bpsc.2018.07.006

Clarke, T.-K., Adams, M., Howard, D., Xia, C., Davies, G., Hayward, C., Campbell, A., Padmanabhan, S., Smith, B., Murray, A., Porteous, D., Deary, I., & McIntosh, A. (2019). Genetic and shared couple environmental contributions to smoking and alcohol use in the UK population. *Molecular Psychiatry*, *N/A*, 1–11. https://doi.org/10.1038/s41380-019-0607-x

Davis, K., Adams, M., Breen, G., Coleman, J., Dregan, A., Gaspar, H., Hubel, C., Lee, W., McIntosh, A., Nolan, J., Pearsall, R., & Hotopf, M. (2019). Indicators of mental disorders in UK Biobank – a comparison of approaches. *International Journal of Methods in Psychiatric Research*. https://doi.org/10.1002/mpr.1796

Frangou, S., Shirali, M., Adams, M., Howard, D., Gibson, J., Hall, L., Smith, B., Padmanabhan, S., Murray, A., Porteous, D., Haley, C., Deary, I., Clarke, T.-K., & McIntosh, A. (2019). Insulin resistance: Genetic associations with depression and cognition in population based cohorts. *Experimental Neurology*, *316*, 20–26. https://doi.org/10.1016/j.expneurol.2019.04.001

Grove, J., Ripke, S., Als, T., Mattheisen, M., Walters, R., Won, H., Pallesen, J., Agerbo, E., Andreassen, O., Anney, R., Awashti, S., Belliveau, R., Bettella, F., Buxbaum, J., Bybjerg-Grauholm, J., Baekvad-Hansen, M., Cerrato, F., Chambert, K., Christensen, J., … 23andMe Research Team. (2019). Identification of common genetic risk variants for autism spectrum disorder. *Nature Genetics*, *51*(3), 431. https://doi.org/10.1038/s41588-019-0344-8

Habota, T., Sandu, A.-L., Waiter, G., Mcneil, C., Steele, J., Macfarlane, J., Whalley, H., Valentine, R., Younie, D., Crouch, N., Hawkins, E., Hirose, Y., Romaniuk, L., Milburn, K., Buchan, G., Coupar, T., Stirling, M., Jagpal, B., Maclennan, B., … McIntosh, A. (2019). Cohort profile for the STratifying Resilience and Depression Longitudinally (STRADL) study: A depression-focused investigation of Generation Scotland, using detailed clinical, cognitive, and neuroimaging assessments. *Wellcome Open Research*, *4*, 185. https://doi.org/10.12688/wellcomeopenres.15538.1

Hafferty, J., Navrady, L., Adams, M., Howard, D., Campbell, A., Whalley, H., Lawrie, S., Nicodemus, K., Porteous, D., Deary, I., & McIntosh, A. (2019). The role of neuroticism in self-harm and suicidal ideation: Results from two UK population-based cohorts. *Social Psychiatry and Psychiatric Epidemiology*, *54*(12), 1505–1518. https://doi.org/10.1007/s00127-019-01725-7

Hafferty, J., Wigmore, E., Howard, D., Adams, M., Clarke, T.-K., Campbell, A., MacIntyre, D., Nicodemus, K., Lawrie, S., Porteous, D., & McIntosh, A. (2019). Pharmaco-epidemiology of antidepressant exposure in a UK cohort record-linkage study. *Journal of Psychopharmacology*, 026988111982788. https://doi.org/10.1177/0269881119827888

Harris, M., Shen, X., Cox, S., Gibson, J., Adams, M., Clarke, T.-K., Deary, I., Lawrie, S., McIntosh, A., & Whalley, H. (2019). Stratifying major depressive disorder by polygenic risk for schizophrenia in relation to structural brain measures. *Psychological Medicine*, *50*(10), 1653–1662. https://doi.org/10.1017/S003329171900165X

Howard, D., Adams, M., Clarke, T.-K., Hafferty, J., Gibson, J., Shirali, M., Coleman, J., Hagenaars, S., Ward, J., Wigmore, E., Alloza, C., Shen, X., Barbu, M., Xu, E., Whalley, H., Marioni, R., Porteous, D., Davies, G., Deary, I., … McIntosh, A. (2019). Genome-wide meta-analysis of depression identifies 102 independent variants and highlights the importance of the prefrontal brain regions. *Nature Neuroscience*, *22*(3), 343–352. https://doi.org/10.1038/s41593-018-0326-7

Johnston, K., Adams, M., Nicholl, B., Ward, J., Strawbridge, R., McIntosh, A., Smith, D., & Bailey, M. (2019). Identification of novel common variants associated with chronic pain using conditional false discovery rate analysis with major depressive disorder and assessment of pleiotropic effects of LRFN5. *Translational Psychiatry*, *9*(1). https://doi.org/10.1038/s41398-019-0613-4

Johnston, K., Adams, M., Nicholl, B., Ward, J., Strawbridge, R., Ferguson, A., McIntosh, A., Bailey, M., & Smith, D. (2019). Genome-wide association study of multisite chronic pain in UK Biobank. *PLoS Genetics*, *15*(6). https://doi.org/10.1371/journal.pgen.1008164

Meng, W., Adams, M., Palmer, C., Shi, J., Auton, A., Ryan, K., Jordan, J., Mitchell, B., Jackson, R., Yau, M., McIntosh, A., Smith, B., & 23andMe Research Team. (2019). Genome-wide association study of knee pain identifies associations with GDF5 and COL27A1 in UK Biobank. *Communications biology*, *2*, 321. https://doi.org/10.1038/s42003-019-0568-2

Neilson, E., Shen, X., Cox, S., Clarke, T.-K., Wigmore, E., Gibson, J., Howard, D., Adams, M., Harris, M., Davies, G., Deary, I., Whalley, H., McIntosh, A., & Lawrie, S. (2019). Impact of polygenic risk for schizophrenia on cortical structure in UK Biobank. *Biological Psychiatry*, *86*(7), 536–544. https://doi.org/10.1016/j.biopsych.2019.04.013

Rosoff, D., Clarke, T.-K., Adams, M., McIntosh, A., Davey Smith, G., Jung, J., & Lohoff, F. (2019). Educational attainment impacts drinking behaviors and risk for alcohol dependence: Results from a two-sample Mendelian randomization study with ≈780,000 participants. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-019-0535-9

Shen, X., Adams, M., Ritakari, T., Cox, S., McIntosh, A., & Whalley, H. (2019). White matter microstructure and its relation to longitudinal measures of depressive symptoms in mid-late life. *Biological Psychiatry*. https://doi.org/10.1016/j.biopsych.2019.06.011

Wigmore, E., Hafferty, J., Hall, L., Howard, D., Clarke, T.-K., Fabbri, C., Lewis, C., Uher, R., Navrady, L., Adams, M., Zeng, Y., Campbell, A., Gibson, J., Thomson, P., Hayward, C., Smith, B., Hocking, L., Padmanabhan, S., Deary, I., … McIntosh, A. (2019). Genome-wide association study of antidepressant treatment resistance in a population-based cohort using health service prescription data and meta-analysis with GENDEP. *Pharmacogenomics Journal*. https://doi.org/10.1038/s41397-019-0067-3

Zeng, Y., Amador, C., Xia, C., Marioni, R., Sproul, D., Walker, R. M., Morris, S. W., Bretherick, A., Canela-Xandri, O., Boutin, T. S., Clark, D. W., Campbell, A., Rawlik, K., Hayward, C., Nagy, R., Tenesa, A., Porteous, D. J., Wilson, J. F., Deary, I. J., … Haley, C. S. (2019). Parent of origin genetic effects on methylation in humans are common and influence complex trait variation. *Nature Communications*, *10*(1), 1383. https://doi.org/10.1038/s41467-019-09301-y

Arnau Soler, A., Adams, M., Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Generation Scotland, Hayward, C., & Thomson, P. (2018). Genome-wide interaction study of a proxy for stress-sensitivity and its prediction of major depressive disorder. *PLoS ONE*, *13*(12). https://doi.org/10.1371/journal.pone.0209160

Clarke, T.-K., Zeng, Y., Navrady, L., Xia, C., Haley, C., Campbell, A., Navarro, P., Amador, C., Adams, M., Howard, D., Soler, A., Hayward, C., Thomson, P., Smith, B., Padmanabhan, S., Hocking, L., Hall, L., Porteous, D., Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, … McIntosh, A. (2018). Genetic and environmental determinants of stressful life events and their overlap with depression and neuroticism. *Wellcome Open Research*, *3*, 11. https://doi.org/10.12688/wellcomeopenres.13893.1

Hall, L., Adams, M., Arnau Soler, A., Clarke, T.-K., Howard, D., Zeng, Y., Davies, G., Hagenaars, S., Fernandez-Pujals, A., Gibson, J., Wigmore, E., Boutin, T., Hayward, C., Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium, Porteous, D., Deary, I., Thomson, P., Haley, C., McIntosh, A., & Generation Scotland. (2018). Genome-wide meta-analyses of stratified depression in Generation Scotland and UK Biobank. *Translational Psychiatry*, *8*. https://doi.org/10.1038/s41398-017-0034-1

Howard, D., Adams, M., Shirali, M., Clarke, T.-K., Marioni, R., Davies, G., Coleman, J., Alloza, C., Shen, X., Barbu, M., Wigmore, E., Gibson, J., 23andMe Research Team, Hagenaars, S., Lewis, C., Ward, J., Smith, D., Sullivan, P., Haley, C., … McIntosh, A. (2018b). Genome-wide association study of depression phenotypes in UK Biobank identifies variants in excitatory synaptic pathways. *Nature Communications*, *9*(1), 1–10. https://doi.org/10.1038/s41467-018-03819-3

Luciano, M., Hagenaars, S., Davies, G., Hill, W., Clarke, T.-K., Shirali, M., Harris, S., Marioni, R., Liewald, D., Fawns-Ritchie, C., Adams, M., Howard, D., Lewis, C., Gale, C., McIntosh, A., & Deary, I. (2018). Association analysis in over 329,000 individuals identifies 116 independent variants influencing neuroticism. *Nature Genetics*, *50*, 6–11. https://doi.org/10.1038/s41588-017-0013-8

Meng, W., Adams, M., Hebert, H., Deary, I., McIntosh, A., & Smith, B. (2018). A genome-wide association study finds genetic associations with broadly-defined headache in UK Biobank (N = 223,773). *EBioMedicine*. https://doi.org/10.1016/j.ebiom.2018.01.023

Navrady, L., Zeng, Y., Clarke, T.-K., Adams, M., Howard, D., Deary, I., & McIntosh, A. (2018). Genetic and environmental contributions to psychological resilience and coping. *Wellcome Open Research*, *3*(12), 12. https://doi.org/10.12688/wellcomeopenres.13854.1

Robinson, L., Coleman, K., Capitanio, J., Gottlieb, D., Handel, I., Adams, M., Leach, M., Warran, N., & Weiss, A. (2018). Rhesus macaque personality, dominance, behavior, and health. *American Journal of Primatology*, *80*(2), 1–8. https://doi.org/10.1002/ajp.22739

Sanchez-Roige, S., Palmer, A., Fontanillas, P., Elson, S., The 23andMe Research Team, Substance Use Disorder Working Group of the Psychiatric Genomics Consortium, Adams, M., Howard, D., Edenberg, H., Davies, G., Crist, R., Deary, I., McIntosh, A., & Clarke, T. (2018). Genome-wide association study meta-analysis of the alcohol use disorders identification test (AUDIT) in two population-based cohorts (n=141,932). *American Journal of Psychiatry*. https://doi.org/10.1176/appi.ajp.2018.18040369

Seltmann, M., Helle, S., Adams, M., Mar, K., & Lahdenperä, M. (2018). Evaluating the personality structure of semi-captive asian elephants living in their natural habitat. *Royal Society Open Science*, *5*(2). https://doi.org/10.1098/rsos.172026

Shen, X., Cox, S., Adams, M., Howard, D., Lawrie, S., Ritchie, S., Bastin, M., Deary, I., McIntosh, A., & Sibley, H. (2018). Resting-state connectivity and its association with cognitive performance, educational attainment, and household income in the UK Biobank. *Biological psychiatry. Cognitive neuroscience and neuroimaging*, *3*(10), 878–886. https://doi.org/10.1016/j.bpsc.2018.06.007

Strawbridge, R., Ward, J., Lyall, L., Tunbridge, E., Cullen, B., Graham, N., Ferguson, A., Johnston, K., Lyall, D., Mackay, D., Cavanagh, J., Howard, D., Adams, M., Deary, I., Escott-Price, V., O'Donovan, M., McIntosh, A., Bailey, M., Pell, J., … Smith, D. (2018). Genetics of self-reported risk-taking behaviour, trans-ethnic consistency and relevance to brain gene expression. *Translational Psychiatry*, *8*(1), 1–11. https://doi.org/10.1038/s41398-018-0236-1

Wray, N., Ripke, S., Mattheisen, M., Trzaskowski, M., Byrne, E., Abdellaoui, A., Adams, M., Agerbo, E., Air, T., Andlauer, T., Bacanu, S.-A., Bækvad-Hansen, M., Beekman, A., Bigdeli, T., Binder, E., Blackwood, D., Bryois, J., Buttenschøn, H., Bybjerg-Grauholm, J., … Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2018). Genome-wide association analyses identify 44 risk variants and refine the genetic architecture of major depression. *Nature Genetics*, *50*(5), 668–681. https://doi.org/10.1038/s41588-018-0090-3

Arnaud, C., Suzumura, T., Inoue, E., Adams, M., Weiss, A., & Inoue-Murayama, M. (2017). Genes, social transmission, but not maternal effects influence responses of wild Japanese macaques *(Macaca fuscata)* to novel-object and novel-food tests. *Primates*, *58*(1), 103–113. https://doi.org/10.1007/s10329-016-0572-9

Clarke, T.-K., Adams, M., Davies, G., Howard, D., Hall, L., Padmanabhan, S., Murray, A., Smith, B., Campbell, A., Hayward, C., Porteous, D., Deary, I., & McIntosh, A. (2017). Genome-wide association study of alcohol consumption and genetic overlap with other health-related traits in UK biobank (n=112 117). *Molecular Psychiatry*, *22*(10), 1376–1384. https://doi.org/10.1038/mp.2017.153

Gibson, J., Russ, T., Adams, M., Clarke, T.-K., Howard, D., Hall, L., Fernandez-Pujals, A., Wigmore, E., Hayward, C., Davies, G., Murray, A., Smith, B., Porteous, D., Deary, I., & McIntosh, A. (2017). Assessing the presence of shared genetic architecture between Alzheimer's disease and major depressive disorder using genome-wide association data. *Translational Psychiatry*, *7*(4). https://doi.org/10.1038/tp.2017.49

Hafferty, J., Campbell, A., Navrady, L., Adams, M., Macintyre, D., Lawrie, S., Nicodemus, K., Porteous, D., & McIntosh, A. (2017). Self-reported medication use validated through record linkage to national prescribing data. *Journal of Clinical Epidemiology*. https://doi.org/10.1016/j.jclinepi.2017.10.013

Howard, D., Adams, M., Clarke, T.-K., Wigmore, E., Zeng, Y., Hagenaars, S., Lyall, D., Thomson, P., Evans, K., Porteous, D., Nagy, R., Hayward, C., Haley, C., Smith, B., Murray, A., Batty, G., Deary, I., & McIntosh, A. (2017). Haplotype-based association analysis of general cognitive ability in Generation Scotland, the English Longitudinal Study of Ageing, and UK Biobank. *Wellcome Open Research*. https://doi.org/10.12688/wellcomeopenres.12171.1

Howard, D., Hall, L., Hafferty, J., Zeng, Y., Adams, M., Clarke, T.-K., Porteous, D., Nagy, R., Hayward, C., Smith, B., Murray, A., Ryan, N., Evans, K., Haley, C., Deary, I., Thomson, P., & McIntosh, A. (2017). Genome-wide haplotype-based association analysis of major depressive disorder in Generation Scotland and UK Biobank. *Translational Psychiatry*, *7*. https://doi.org/10.1038/s41398-017-0010-9

Navrady, L., Adams, M., Chan, W., Ritchie, S., McIntosh, A., & Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2017). Genetic risk of major depressive disorder: The moderating and mediating effects of neuroticism and psychological resilience on clinical and self-reported depression. *Psychological Medicine*. https://doi.org/10.1017/S0033291717003415

Navrady, L., Ritchie, S., Chan, S., Kerr, D., Adams, M., Hawkins, E., Porteous, D., Deary, I., Gale, C., Batty, G., & McIntosh, A. (2017). Intelligence and neuroticism in relation to depression and psychological distress: Evidence from two large population cohorts. *European Psychiatry*, *43*, 58–65. https://doi.org/10.1016/j.eurpsy.2016.12.012

Reus, L., Shen, X., Gibson, J., Wigmore, E., Ligthart, L., Adams, M., Davies, G., Cox, S., Hagenaars, S., Bastin, M., Deary, I., Sibley, H., & McIntosh, A. (2017). Association of polygenic risk for major psychiatric illness with subcortical volumes and white matter integrity in UK Biobank. *Scientific Reports*, *7*. https://doi.org/10.1038/srep42140

Shen, X., Reus, L., Cox, S., Adams, M., Liewald, D., Bastin, M., Smith, D., Deary, I., Sibley, H., & McIntosh, A. (2017). Subcortical volume and white matter integrity abnormalities in major depressive disorder: Findings from UK Biobank imaging data. *Scientific Reports*, *7*(1), 5547. https://doi.org/10.1038/s41598-017-05507-6

Wigmore, E., Clarke, T.-K., Howard, D., Adams, M., Hall, L., Zeng, Y., Gibson, J., Davies, G., Fernandez-Pujals, A., Thomson, P., Hayward, C., Smith, B., Hocking, L., Padmanabhan, S., Deary, I., Porteous, D., Nicodemus, K., & McIntosh, A. (2017). Do regional brain volumes and major depressive disorder share genetic architecture?: A study of Generation Scotland (n=19,762), UK Biobank (n=24,048) and the English Longitudinal Study of Ageing (n=5,766). *Translational Psychiatry*, *7*(8). https://doi.org/10.1038/tp.2017.148

Zeng, Y., Navarro, P., Shirali, M., Howard, D., Adams, M., Hall, L., Clarke, T.-K., Thomson, P., Smith, B., Murray, A., Padmanabhan, S., Hayward, C., Boutin, T., MacIntyre, D., Lewis, C., Wray, N., Mehta, D., Penninx, B., Milaneschi, Y., … McIntosh, A. (2017). Genome-wide regional heritability mapping identifies a locus within the TOX2 gene associated with major depressive disorder. *Biological Psychiatry*, *82*(5), 312–321. https://doi.org/10.1016/j.biopsych.2016.12.012

Sibley, H., Adams, M., Hall, L., Clarke, T.-K., Fernandez-Pujals, A., Gibson, J., Wigmore, E., Hafferty, J., Hagenaars, S., Davies, G., Campbell, A., Hayward, C., Lawrie, S., Porteous, D., Deary, I., & McIntosh, A. (2016). Dissection of major depressive disorder using polygenic risk scores for schizophrenia in two independent cohorts. *Translational Psychiatry*, *6*(11), e938. https://doi.org/10.1038/tp.2016.207

Smith, D., Escott-Price, V., Davies, G., Bailey, M., Colodro-Conde, L., Ward, J., Vedernikov, A., Marioni, R., Cullen, B., Lyall, D., Hagenaars, S., Liewald, D., Luciano, M., Gale, C., Ritchie, S., Hayward, C., Nicholl, B., Bulik-Sullivan, B., Adams, M., … O'Donovan, M. (2016). Genome-wide analysis of over 106 000 individuals identifies 9 neuroticism-associated loci. *Molecular Psychiatry*, *21*(6), 749–757. https://doi.org/10.1038/mp.2016.49

Fernandez-Pujals, A., Adams, M., Thomson, P., McKechanie, A., Blackwood, D., Smith, B., Dominiczak, A., Morris, A., Matthews, K., Campbell, A., Linksted, P., Haley, C., Deary, I., Porteous, D., MacIntyre, D., & McIntosh, A. (2015). Epidemiology and heritability of major depressive disorder, stratified by age of onset, sex, and illness course in Generation Scotland: Scottish Family Health Study (GS:SFHS). *PLoS ONE*, *10*(11), e0142197. https://doi.org/10.1371/journal.pone.0142197

Robinson, L., Coleman, K., Gottlieb, D., Adams, M., Handel, I., Leach, M., Waran, N., & Weiss, A. (2015). Testing a shortened personality questionnaire with captive rhesus macaques *(Macaca mulatta)* at the Oregon National Primate Research Center. *American Journal of Primatology*, *77*, 142–142.

Weiss, A., Adams, M., & King, J. (2011). Happy orang-utans live longer lives. *Biology Letters*, *7*(6), 872–874. https://doi.org/10.1098/rsbl.2011.0543

Weiss, A., Adams, M., & Johnson, W. (2011). The Big None: No evidence for a general factor of personality in chimpanzees, orangutans, or rhesus macaques. *Journal of Research in Personality*, *45*(4), 393–397. https://doi.org/10.1016/j.jrp.2011.04.006

Wilson, A., Morrissey, M., Adams, M., Walling, C., Guinness, F., Pemberton, J., Clutton-Brock, T., & Kruuk, L. (2011). Indirect genetics effects and evolutionary constraint: An analysis of social dominance in red deer, cervus elaphus. *Journal of Evolutionary Biology*, *24*(4), 772–783. https://doi.org/10.1111/j.1420-9101.2010.02212.x

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*Contributed data to secondary analysis or as a consortium member.*

Koromina, M., Ravi, A., Panagiotaropoulou, G., Schilder, B. M., Humphrey, J., Braun, A., Bidgeli, T., Chatzinakos, C., Coombes, B. J., Kim, J., Liu, X., Terao, C., O'Connell, K. S., Adams, M. J., Adolfsson, R., Alda, M., Alfredsson, L., Andlauer, T. F. M., Andreassen, O. A., … Mullins, N. (2025). Fine-mapping genomic loci refines bipolar disorder risk genes. *Nature Neuroscience*. https://doi.org/10.1038/s41593-025-01998-z

Meng, X., Navoly, G., Giannakopoulou, O., Levey, D. F., Koller, D., Pathak, G. A., Koen, N., Lin, K., Adams, M. J., Rentería, M. E., Feng, Y., Gaziano, J. M., Stein, D. J., Zar, H. J., Campbell, M. L., van Heel, D. A., Trivedi, B., Finer, S., McQuillin, A., … Kuchenbaecker, K. (2024). Multi-ancestry genome-wide association study of major depression aids locus discovery, fine mapping, gene prioritization and causal inference. *Nature Genetics*. https://doi.org/10.1038/s41588-023-01596-4

Docherty, A. R., Mullins, N., Ashley-Koch, A. E., Qin, X., Coleman, J. R., Shabalin, A., Kang, J., Murnyak, B., Wendt, F., Adams, M., Campos, A. I., DiBlasi, E., Fullerton, J. M., Kranzler, H. R., Bakian, A. V., Monson, E. T., Rentería, M. E., Walss-Bass, C., Andreassen, O. A., … Ruderfer, D. M. (2023). GWAS meta-analysis of suicide attempt: Identification of 12 genome-wide significant loci and implication of genetic risks for specific health factors. *American Journal of Psychiatry*, *180*(10), 723–738. https://doi.org/10.1176/appi.ajp.21121266

Guintivano, J., Byrne, E. M., Kiewa, J., Yao, S., Bauer, A. E., Aberg, K. A., Adams, M. J., Campbell, A., Campbell, M. L., Choi, K. W., Corfield, E. C., Havdahl, A., Hucks, D., Koen, N., Lu, Y., Mægbæk, M. L., Mullaert, J., Peterson, R. E., Raffield, L. M., … Sullivan, P. (2023). Meta-analyses of genome-wide association studies for postpartum depression. *American Journal of Psychiatry*, *180*(12), 884–895. https://doi.org/10.1176/appi.ajp.20230053

Kimbrel, N. A., Ashley-Koch, A. E., Qin, X. J., Lindquist, J. H., Garrett, M. E., Dennis, M. F., Hair, L. P., Huffman, J. E., Jacobson, D. A., Madduri, R. K., Trafton, J. A., Coon, H., Docherty, A. R., Mullins, N., Ruderfer, D. M., Harvey, P. D., McMahon, B. H., Oslin, D. W., Beckham, J. C., … Million Veteran Program Suicide Exemplar Workgroup, the International Suicide Genetics Consortium, the Veterans Affairs Mid-Atlantic Mental Illness Research, Education, and Clinical Center Workgroup, and the Veterans Affairs Million Veteran Program. (2023). Identification of novel, replicable genetic risk loci for suicidal thoughts and behaviors among us military veterans. *JAMA Psychiatry*, *80*(2), 135–145. https://doi.org/10.1001/jamapsychiatry.2022.3896

Mullins, N., Kang, J., Campos, A. I., Coleman, J. R., Edwards, A. C., Galfalvy, H., Levey, D. F., Lori, A., Shabalin, A., Starnawska, A., Su, M.-H., Watson, H. J., Adams, M., Awasthi, S., Gandal, M., Hafferty, J. D., Hishimoto, A., Kim, M., Okazaki, S., … Ruderfer, D. M. (2022). Dissecting the shared genetic architecture of suicide attempt, psychiatric disorders, and known risk factors. *Biological Psychiatry*, *91*(3), 313–327. https://doi.org/10.1016/j.biopsych.2021.05.029

Trubetskoy, V., Pardiñas, A. F., Qi, T., Panagiotaropoulou, G., Awasthi, S., Bigdeli, T. B., Bryois, J., Chen, C.-Y., Dennison, C. A., Hall, L. S., Lam, M., Watanabe, K., Frei, O., Ge, T., Harwood, J. C., Koopmans, F., Magnusson, S., Richards, A. L., Sidorenko, J., … Bertolino, A. (2022). Mapping genomic loci implicates genes and synaptic biology in schizophrenia. *Nature*. https://doi.org/10.1038/s41586-022-04434-5

Blokland, G., Grove, J., Chen, C.-Y., Cotsapas, C., Tobet, S., Handa, R., St Clair, D., Lencz, T., Mowry, B., Periyasamy, S., Cairns, M., Tooney, P., Wu, J., Kelly, B., Kirov, G., Sullivan, P., Corvin, A., Riley, B., Esko, T., … Schizophrenia Working Group of the Psychiatric Genomics Consortium. (2021). Sex-dependent shared and nonshared genetic architecture across mood and psychotic disorders. *Biological Psychiatry*. https://doi.org/10.1016/j.biopsych.2021.02.972

Campos, A., Thompson, P., Veltman, D., Pozzi, E., Van Veltzen, L., Jahanshad, N., Adams, M., Baune, B., Berger, K., Brosch, K., Bülow, R., Connolly, C., Dannlowski, U., Davey, C., De Zubicaray, G., Dima, D., Erwin-grabner, T., Evans, J., Fu, C., … ENIGMA-Major Depressive Disorder Working Group. (2021). Brain correlates of suicide attempt in 18,925 participants across 18 international cohorts. *Biological Psychiatry*. https://doi.org/10.1016/j.biopsych.2021.03.015

Giannakopoulou, O., Lin, K., Meng, X., Su, M.-h., Kuo, P.-h., Peterson, R., Awasthi, S., Moscati, A., Coleman, J., Bass, N., Millwood, I., Chen, Y., Chen, Z., Chen, H.-c., Lu, M.-l., Huang, M.-c., Chen, C.-h., Stahl, E., Loos, R., … 23andMe Research Team. (2021). The genetic architecture of depression in individuals of East Asian ancestry: A genome-wide association study. *JAMA Psychiatry*. https://doi.org/10.1001/jamapsychiatry.2021.2099

Kalman, J., Olde Loohuis, L., Vreeker, A., McQuillin, A., Stahl, E., Ruderfer, D., Grigoroiu-Serbanescu, M., Panagiotaropoulou, G., Ripke, S., Bigdeli, T., Stein, F., Meller, T., Meinert, S., Pelin, H., Streit, F., Papiol, S., Adams, M., Adolfsson, R., Adorjan, K., … Psychiatric Genomics Consortium (PGC) Bipolar Disorder Working Group, International Consortium on Lithium Genetics (ConLiGen), Colombia-US Cross Disorder Collaboration in Psychiatric Genetics. (2021). Characterisation of age and polarity at onset in bipolar disorder. *The British Journal of Psychiatry*. https://doi.org/10.1192/bjp.2021.102

Miles, A., Dos Santos, F., Byrne, E., Renteria, M., McIntosh, A., Adams, M., Pistis, G., Castelao, E., Preisig, M., Baune, B., Schubert, K., Lewis, C., Jones, L., Jones, I., Uher, R., Smoller, J., Perlis, R., Levinson, D., Potash, J., … Nikolova, Y. (2021). Transcriptome-based polygenic score links depression-related corticolimbic gene expression changes to sex-specific brain morphology and depression risk. *Neuropsychopharmacology*. https://doi.org/10.1038/s41386-021-01189-x

Sewell, M., Shen, X., Jiménez-Sánchez, L., Edmondson-Stait, A., Green, C., Adams, M., McIntosh, A., Lyall, D., Whalley, H., & Lawrie, S. (2021). Associations between major psychiatric disorder polygenic risk scores and blood-based markers in UK Biobank. *Brain, Behavior, and Immunity*. https://doi.org/10.1016/j.bbi.2021.06.002

Cai, N., Revez, J., Adams, M., Andlauer, T., Breen, G., Byrne, E., Clarke, T.-K., Forstner, A., Grabe, H., Hamilton, S., Levinson, D., Lewis, C., Lewis, G., Martin, N., Milaneschi, Y., Mors, O., Müller-myhsok, B., Penninx, B., Perlis, R., … Flint, J. (2020). Minimal phenotyping yields genome-wide association signals of low specificity for major depression. *Nature Genetics*, *52*(4), 437–447. https://doi.org/10.1038/s41588-020-0594-5

Coleman, J. R. I., Peyrot, W. J., Purves, K. L., Davis, K. A. S., Rayner, C., Choi, S. W., Hübel, C., Gaspar, H. A., Kan, C., Van der Auwera, S., Adams, M. J., Lyall, D. M., Choi, K. W., Wray, N. R., Ripke, S., Mattheisen, M., Trzaskowski, M., Byrne, E. M., Abdellaoui, A., … Major Depressive Disorder Working Group the Psychiatric Genomics Consortium. (2020). Genome-wide gene-environment analyses of major depressive disorder and reported lifetime traumatic experiences in UK Biobank. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-019-0546-6

Coleman, J., Gaspar, H., Bryois, J., Byrne, E., Forstner, A., Holmans, P., de Leeuw, C., Mattheisen, M., McQuillin, A., Whitehead Pavlides, J., Pers, T., Ripke, S., Stahl, E., Steinberg, S., Trubetskoy, V., Trzaskowski, M., Wang, Y., Abbott, L., Abdellaoui, A., … Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2020). The genetics of the mood disorder spectrum: Genome-wide association analyses of more than 185,000 cases and 439,000 controls. *Biological Psychiatry*. https://doi.org/10.1016/j.biopsych.2019.10.015

Hagenaars, S., Coleman, J., Choi, S., Gaspar, H., Adams, M., Howard, D., Hodgson, K., Traylor, M., Air, T., Andlauer, T., Arolt, V., Baune, B., Binder, E., Blackwood, D., Boomsma, D., Campbell, A., Cearns, M., Czamara, D., Dannlowski, U., … Lewis, C. (2020). Genetic comorbidity between major depression and cardio-metabolic traits, stratified by age at onset of major depression. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*. https://doi.org/10.1002/ajmg.b.32807

Johnson, E., Demontis, D., Thorgeirsson, T., Walters, R., Polimanti, R., Hatoum, A., Sanchez-roige, S., Paul, S., Wendt, F., Clarke, T.-K., Lai, D., Reginsson, G., Zhou, H., He, J., Baranger, D., Gudbjartsson, D., Wedow, R., Adkins, D., Adkins, A., … Agrawal, A. (2020). A large-scale genome-wide association study meta-analysis of cannabis use disorder. *The Lancet Psychiatry*, *7*(12), 1032–1045. https://doi.org/10.1016/S2215-0366(20)30339-4

Johnson, E., Sanchez-Roige, S., Acion, L., Adams, M., Bucholz, K., Chan, G., Chao, M., Chorlian, D., Dick, D., Edenberg, H., Foroud, T., Hayward, C., Heron, J., Hesselbrock, V., Hickman, M., Kendler, K., Kinreich, S., Kramer, J., Kuo, S., … Edwards, A. (2020). Polygenic contributions to alcohol use and alcohol use disorders across population-based and clinically ascertained samples. *Psychological Medicine*, 1–10. https://doi.org/10.1017/S0033291719004045

Lohoff, F., Roy, A., Jung, J., Longley, M., Rosoff, D., Luo, A., O'connell, E., Sorcher, J., Sun, H., Schwandt, M., Hodgkinson, C., Goldman, D., Momenan, R., McIntosh, A., Adams, M., Walker, R., Evans, K., Porteous, D., Smith, A., … Kaminsky, Z. (2020). Epigenome-wide association study and multi-tissue replication of individuals with alcohol use disorder: Evidence for abnormal glucocorticoid signaling pathway gene regulation. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-020-0734-4

McEwan, A., Davidson, C., Hay, E., Turnbull, Y., Erickson, J., Marini, P., Wilson, D., McIntosh, A., Adams, M., Murgatroyd, C., Barrett, P., Delibegovic, M., Clarke, T.-K., & MacKenzie, A. (2020). CRISPR disruption and UK Biobank analysis of a highly conserved polymorphic enhancer suggests a role in male anxiety and ethanol intake. *Molecular Psychiatry*, *2020*, 1–14. https://doi.org/10.1038/s41380-020-0707-7

Bermingham, M., Walker, R., Marioni, R., Morris, S., Rawlik, K., Zeng, Y., Campbell, A., Redmond, P., Whalley, H., Adams, M., Hayward, C., Deary, I., Porteous, D., McIntosh, A., & Evans, K. (2019). Identification of novel differentially methylated sites with potential as clinical predictors of impaired respiratory function and COPD. *EBioMedicine*, *43*, 576–586. https://doi.org/10.1016/j.ebiom.2019.03.072

Choi, K., Chen, C., Stein, M., Klimentidis, Y., Wang, M., Koenen, K., Smoller, J., Wray, N., Ripke, S., Mattheisen, M., Trzaskowski, M., Byrne, E., Abdellaoui, A., Adams, M., Agerbo, E., Air, T., Andlauer, T., Bacanu, S., Bækvad-Hansen, M., . . . Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2019). Assessment of bidirectional relationships between physical activity and depression among adults a 2-sample Mendelian randomization study. *JAMA Psychiatry*, *76*(4), 399–408. https://doi.org/10.1001/jamapsychiatry.2018.4175

Czamara, D., Eraslan, G., Page, C., Lahti, J., Lahti-Pulkkinen, M., Hämäläinen, E., Kajantie, E., Laivuori, H., Villa, P., Reynolds, R., Nystad, W., Håberg, S., London, S., O'Donnell, K., Garg, E., Meaney, M., Entringer, S., Wadhwa, P., Buss, C., . . . Consortium, M. (2019). Integrated analysis of environmental and genetic influences on cord blood DNA methylation in new-borns. *Nature Communications*, *10*(1). https://doi.org/10.1038/s41467-019-10461-0

Walters, R., Polimanti, R., Johnson, E., Mcclintick, J., Adams, M., Adkins, A., Aliev, F., Bacanu, S.-A., Batzler, A., Bertelsen, S., Biernacka, J., Bigdeli, T., Chen, L.-s., Clarke, T.-K., Chou, Y.-l., Degenhardt, F., Docherty, A., Edwards, A., Fontanillas, P., . . . Agrawal, A. (2018). Transancestral GWAS of alcohol dependence reveals common genetic underpinnings with psychiatric disorders. *Nature Neuroscience*, *21*(12), 1656–1669. https://doi.org/10.1038/s41593-018-0275-1

Marioni, R., Ritchie, S., Joshi, P., Hagenaars, S., Okbay, A., Fischer, K., Adams, M., Hill, W., Davies, G., Nagy, R., Amador, C., Läll, K., Metspalu, A., Liewald, D., Campbell, A., Wilson, J., Hayward, C., Esko, T., Porteous, D., . . . Gale, C. (2016). Genetic variants linked to education predict longevity. *Proceedings of the National Academy of Sciences*, *113*(47), 13366–13371. https://doi.org/10.1073/pnas.1605334113

### Book Chapters

Weiss, A., & Adams, M. (2013). Differential behavioral ecology. In C. Carere & D. Maestripieri (Eds.), *Animal personalities: Behavior, physiology, and evolution* (pp. 94–123). The University of Chicago Press.

Adams, M. (2011). Evolutionary genetics of personality in nonhuman primates. In M. Inoue-Murayama, A. Weiss, & S. Kawamura (Eds.), *From genes to animal behavior: Social structures, personality, communication by color* (pp. 137–164). Springer. https://doi.org/10.1007/978-4-431-53892-9_6

Weiss, A., & Adams, M. (2010). Personality, temperament, and behavioral syndromes. In G. Koob, M. Le Moal, & R. Thompson (Eds.), *Encyclopedia of behavioral neuroscience* (3rd ed., pp. 47–53). Elsevier. https://doi.org/10.1016/B978-0-08-045396-5.00117-2

### Invited talks

| | |
|---|---|
| 2019 | College of Medicine, University of Malawi |
| 2014 | Institute of Psychiatry, King's College London |
| | Department of Psychology, University of Sheffield |
| | Department of Psychology, University of York |
| 2013 | German Primate Center. Göttingen |

## Teaching

### Undergraduate

Genetic and Environmental Influences on Behaviour and Mental Health
Data Science for Health and Biomedical Sciences
Medical Science 4 seminar on "Psychiatric Genetics"
Psychology 2 Practical Course (tutor)
Psychology 3 Methods (tutor)

### Postgraduate

Univariate and Multivariate Statistics with R

### Workshops

GenomicSEM Module, International Statistical Genetics Workshop

## Supervision

### Doctoral students

| | |
|---|---|
| 2021 | Keira Johnston, Precision Medicine |
| 2023 | Melissa Lewins, Precision Medicine |
| 2025 | Ella Davyson, Biomedical Artificial Intelligence |

### MSc & Honours Students

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|---|---|
| U of Edinburgh | Quantitative Genetics & Genome Analysis |
| | Biomedical Artificial Intelligence |
| | Integrative Neuroscience |
| | Biomedical Sciences and Medical Sciences |
| U of Sheffield | Sheffield Undergraduate Research Experience |

## Knowledge Exchange

| | |
|---|---|
| 2025 | Generation Scotland Symposium |
| | Usher Building Showcase Exhibition |
| 2024 | Doors Open Day |
| 2023 | Using de-identified Scottish primary care data for research: A meeting of stakeholders |
| 2021 | Evidence Submitted to Cross Party Group – Mental Health, Scottish Parliament |
| | *Depression Detectives*, online participant-led research forum |
| 2020 | Consultation on Digital Strategy for Scotland, Scottish Government |
| | NHS Research Scotland Mental Health Annual Scientific Meeting |
| 2019 | Midlothian Science Festival |
| | HDR UK Schools Engagement |
| | Edinburgh International Science Festival |

## Consulting & Industry Collaborations

23andMe
Baillie Gifford
University of Cambridge

## Open Science

### Code repositories

| | |
|---|---|
| Personal | `https://github.com/mja` |
| Group | `https://github.com/ccbs-stradl` |
| Consortium | `https://github.com/psychiatric-genomics-consortium` |

### Datasets

Adams, M. J., Lewis, C., McIntosh, A., & Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2025). GWAS summary statistics for major depression (PGC MDD2025). Dataset. https://doi.org/10.6084/m9.figshare.27061255

Adams, M. J., & Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2025). MDD2 (MDD2018) GWAS sumstats w/o UKBB. Dataset. https://doi.org/10.6084/m9.figshare.21655784

Adams, M. J., & Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. (2024). GWAS summary statistics for major depressive disorder symptoms (2024). Dataset. https://doi.org/10.6084/m9.figshare.22745573

Wigmore, E., Hafferty, J., Hall, L., Howard, D., Clarke, T.-K., Fabbri, C., Lewis, C., Uher, R., Navrady, L., Adams, M., Zeng, Y., Campbell, A., Gibson, J., Thomson, P., Hayward, C., Smith, B., Hocking, L., Padmanabhan, S., Deary, I., … McIntosh, A. (2020). Genome-wide association summary statistics of antidepressant treatment resistance meta-analysis, [dataset]. https://doi.org/10.7488/ds/2791

Adams, M., McIntosh, A., Campbell, A., Dashti, H., Hill, W. D., Howard, D., Davis, K., Clarke, T.-K., Deary, I., Hayward, C., Porteous, D., & Hotopf, M. (2019). Factors associated with sharing email information and mental health survey participation in large population cohorts, 2006-2017 [dataset]. https://doi.org/10.7488/ds/2554

Howard, D., Adams, M., Clarke, T.-K., Hafferty, J., Gibson, J., Shirali, M., Coleman, J., Ward, J., Wigmore, E., Alloza, C., Shen, X., Barbu, M., Xu, E., Whalley, H., Marioni, R., Porteous, D., Davies, G., Deary, I., Hemani, G., … McIntosh, A. (2019). Genome-wide meta-analysis of depression identifies 102 independent variants and highlights the importance of the prefrontal brain regions, [dataset]. https://doi.org/10.7488/ds/2458

Howard, D., Adams, M., Shirali, M., Clarke, T.-K., Marioni, R., Davies, G., Coleman, J., Alloza, C., Shen, X., Barbu, M., Wigmore, E., Gibson, J., 23andMe Research Team, Hagenaars, S., Lewis, C., Ward, J., Smith, D., Sullivan, P., Haley, C., … McIntosh, A. (2018a). Summary statistics for three depression phenotypes in UK Biobank, [dataset]. https://doi.org/10.7488/ds/2350

Adams, M. J., Robinson, M. R., Mannarelli, M.-E., & Hatchwell, B. (2015b). Data from: Social genetic and social environment effects on parental and helper care in a cooperatively breeding bird. https://doi.org/10.5061/dryad.9nj6s

# Journals, Conferences, Grant Panels

## Editorial Boards

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|---|---|
| 2014–2016 | Frontiers in Evolutionary and Population Genetics |

## Reviewer

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|---|---|
| Journals | American Journal of Primatology, Animal Behaviour, BJPsychOpen, Behavioral Ecology and Sociobiology, Biological Psychiatry, Evolution, Intelligence, International Journal of Primatology, JAMA Psychiatry, Journal of Comparative Psychology, Journal of Research in Personality, Journal of the American Academy of Child and Adolescent Psychiatry, Lancet Psychiatry, Molecular Psychiatry, Proceedings of the Royal Society B, Psychological Science |
| Conferences | World Congress of Psychiatric Genetics |
| Grants | Medical Research Council, Swiss National Science Foundation |